
Bridging the Gap Between Scientific Laws Derived by AI Systems and Canonical Knowledge via Abductive Inference with AI-Noether

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Abstract

Advances in AI have shown great potential in contributing to the acceleration of scientific discovery. Symbolic regression can fit interpretable models to data, but these models are not necessarily derivable from established theory. Recent systems (e.g., AI-Descartes, AI-Hilbert) enforce derivability from prior knowledge. However, when existing theories are incomplete or incorrect, these machine-generated hypotheses may fall outside the theoretical scope. Automatically finding corrections to axiom systems to close this gap remains a central challenge in scientific discovery. We propose a solution that evolves the abductive theory-completion step of the scientific method: an open-source algebraic geometry-based system that, given an incomplete axiom system expressible as polynomials and a hypothesis that the axioms cannot derive, generates a set of candidate axioms that, when added to the theory, provably derive the (possibly noisy) hypothesis. We illustrate the efficacy of our approach by showing that it can reconstruct key axioms required to derive the carrier-resolved photo-Hall effect, Einstein’s relativistic laws, and several other laws.

Introduction

The proliferation of artificial intelligence in scientific discovery allows us to extend our exploration of science while also bringing a fundamental question to the forefront: How can we integrate AI-generated insights into canonical scientific knowledge? Since the release of ChatGPT in November 2022, the ubiquity of generative AI has made it easier and cheaper to make scientific discoveries [2, 22, 14, 48, 6, 26, 27, 18, 4, 50, 5, 34, 41]. Indeed, machine learning and AI techniques have

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recently developed new antivenom [49], sorting algorithms [35], drugs that have progressed to phase II clinical trials [53], mathematical discoveries [42, 24], and equations in psychology [40], to name a few applications. More recently, there has even been progress in integrating data and background theory within the scientific discovery process using symbolic techniques [9, 11].

However, the acceleration of scientific discovery poses both challenges and opportunities for the scientific community. First, the reduced cost of making and disseminating incremental scientific discoveries in the age of AI, when combined with strong incentives for scientists to publish-or-perish [39], threatens to overwhelm the peer-review system and obscure genuine advances with incremental contributions. Indeed, LLM use in writing arXiv papers has increased from negligible levels in 2022 to widespread adoption by 2026 [20], with greater use among authors who post preprints more frequently [29]. Second, from a scientific accuracy perspective, it is often unclear whether scientific discoveries made with generative AI or machine learning generalize well to unseen data and advance understanding [25, 37], or merely fit known data well. This lack of interpretability and generalizability (see [30, 43] for discussions) means that verifying scientific discoveries (e.g., via peer review) is becoming as important as the discovery process itself [10]. Third, stacking AI-driven discoveries on top of each other poses a serious risk in domains where absolute truth matters more than predictive accuracy. This is especially true for discoveries generated by AI systems that prioritize minimizing in-sample training loss and may not generalize out-of-distribution. In such an environment, this can result in a scientific model collapse in the sense of [46] where AI-systems stack plausible-seeming but incorrect hallucinations on top of one another, similar to the replication crisis that struck psychology in the early 2010s [45].

To address the challenges posed by and capitalize on the opportunities presented by AI-driven scientific discovery, we require a framework for systematic revision of scientific foundations - what Kuhn called paradigm shifts [28]. In the life cycle of a scientific paradigm, a scientific community starts with a certain set of natural laws and beliefs (axioms). They then use these axioms and experimental data to make scientific discoveries, built on their axiomatic foundation, until the gap between the foundation and prominent discoveries becomes so apparent that a paradigm shift to new axioms is needed. Kolmogorov’s axioms in probability theory [21] are a good example of this process. Before Kolmogorov introduced his axioms in 1933, discoveries were made in an ad hoc manner, leading to paradoxes such as Bertrand’s paradox [44]. However, following Kolmogorov’s paradigm shift, it became possible to distinguish between correct and incorrect discoveries in probability theory. While the natural sciences involve more empirical complexities, Kuhnian paradigm shifts regularly occur, leading to significant breakthroughs when they do. For instance, to develop his theory of special relativity, Einstein had to modify fundamental postulates of prior Newtonian mechanics, such as recognizing that the speed of light is constant [7]. Similarly, Maxwell’s revision of Ampère’s circuital law to include a displacement term completed classical electrodynamics for time-varying fields and led to his prediction that light is an electromagnetic wave [32].

These observations, coupled with the increasing availability of computational power driven by Moore’s Law [33], suggest that AI and machine learning should be deployed to develop new scientific paradigms, much as AI has successfully been applied to scientific discovery. In particular, abductive inference should be employed within the scientific method to determine when a new scientific paradigm is necessary and when the set of scientific axioms used by a community needs revision. Accordingly, we propose an AI framework to solve the problem of abductive reasoning.

We present AI-Noether, an abductive inference system that identifies missing axioms in scientific theories. Given background axioms and a hypothesis that cannot be derived from them, AI-Noether systematically generates candidate axioms whose addition makes the hypothesis derivable, transforming paradigm shifts that were historically informal and human-driven into a scalable computational procedure. This enables us not only to identify discrepancies between theory and candidate hypotheses, but to resolve them by inferring the axiomatic modifications required. More broadly, AI-Noether introduces a complementary discovery mode, in which scientific progress arises not from generating a new empirical law, but from abductively discovering the theoretical axioms that explain why such a law should hold. Figure 1 illustrates that AI-Noether complements traditional discovery by using generated hypotheses and known axioms to discover new axioms essential to explain hypotheses. Figure 2 illustrates in more detail how AI-Noether is uniquely and independently positioned in the broader context of systems for scientific discovery; we expand on this view in Supplementary Note 1.

We name our proposal with an AI- prefix for two reasons. First, it executes abductive inference, a type of reasoning which is known to be computationally challenging to perform [16, 17], and thus fits within the original definition of artificial intelligence proposed by John McCarthy and colleagues in 1955 [36]. Second, our proposal automates a distinct step of the scientific method, and hence fits within a broader end-to-end pipeline alongside other AI- techniques that could eventually automate the entire scientific method [48, 9, 11].

We name our system AI-Noether in honor of Emmy Noether, one of the leading mathematicians of her time who made significant contributions to both algebraic geometry and physics. We build on her work at the core of our system in order to make this form of abductive inference computationally tractable. This name also reflects Noether’s legacy of uncovering the algebraic structures underlying physical laws, most famously via Noether’s theorem on symmetries and conserved quantities [38], paralleling our goal of inferring additional axioms needed to render a candidate hypothesis derivable. To support reproducibility and future research, we open-source AI-Noether’s complete implementation, including all benchmark problems used in this study.

To the best of our knowledge, this is the first research work to study automated abductive reasoning in the context of scientific discovery, in which derivations may consist of fully general polynomial axioms over any set of variables; see Supplementary Note 1 for a review of existing automated approaches to abductive inference and a comparison of our system with existing AI-driven approaches to scientific discovery of mathematical models and Supplementary Note 3 for the technical details, definitions and relevant theorems of this work. Thus, our work constitutes a step toward the larger goal of augmenting and transforming the entire scientific method using AI by bridging the gap between machine-generated hypotheses and known theory.

Results

In the pursuit of scientific discovery, we often find ourselves confronted with a challenge: established theoretical frameworks routinely fall short of fully explaining observable phenomena. For instance, standard models of the Big Bang theory predict that there should be about three times as much lithium in the universe as is observed [19]. This disconnect between theory and observation presents an important question: how do we systematically identify what is missing from our understanding?

Resolving discrepancies between theory and observation is increasingly relevant in the era of machine learning, because data-driven methods can reveal patterns that existing theory cannot explain. Thus, fully integrating insights from statistical AI into science is of considerable interest. However, discoveries from statistical AI are often difficult to verify due to their uninterpretable nature [43, 30], and integrating them into science remains an open problem. Bridging this disconnect can be expressed as the following abductive inference problem: given a set of known scientific principles (axioms) and a newly discovered relationship that cannot be derived from them, what additional principles must we postulate to bridge the gap?

In this section, we formalize this problem for the case where both axioms and discoveries can be expressed as multivariate polynomial equations. We assume that all axioms, including any missing ones, and the hypotheses, are expressible as polynomial equations of the form $A(\mathbf{x}) = 0$. A collection of polynomial equations $A_1(\mathbf{x}) = 0, \dots, A_k(\mathbf{x}) = 0$ implies the polynomial equation $Q(\mathbf{x}) = 0$ if $Q(\bar{\mathbf{x}}) = 0$ for every solution $\bar{\mathbf{x}}$ of the system $\{A_1, \dots, A_k\}$.

Formally, we consider the problem: given axioms $A_1, \dots, A_k$ and a hypothesis Q such that $A_1, \dots, A_k$ do not imply Q , find additional r axioms $A_{k+1}, \dots, A_{k+r}$ with the smallest residual relative to $A_1, \dots, A_k$ such that the augmented system $A_1, \dots, A_{k+r}$ implies Q .

Throughout the rest of this work, we say synonymously that $A_1, \dots, A_k$ explain, derive, or imply Q , or that Q is a consequence of $A_1, \dots, A_k$. Intuitively, the notion of smallest residual corresponds to the idea that we wish to add as little new information as possible to $A_1, \dots, A_k$ to derive Q ; we make this notion precise in the AI-Noether Pipeline subsection below and in Supplementary Notes 2 and 3. We assume that no data are available for the missing axioms $A_{k+1}, \dots, A_{k+r}$; otherwise, abduction would be equivalent to hypothesis generation, a topic that has been frequently studied in scientific discovery. Under this terminology, our system generates a set of candidate axioms $\{\hat{A}_{k+1}, \dots, \hat{A}_l\}$ that, along with the known axioms, explain Q . Further, if no single candidate axiom can be added to explain Q , our system detects this. Current machine-assisted discovery tools either rely on partial or complete background theory, or none at all, to discover mathematical models that fit observational

data, but few append or correct the theory in an automated way, and none do so at the same level of generality. We will illustrate the use of these terms and results with real examples in the Methods, but see Supplementary Note 2 for a synthetic walkthrough example.

In the remainder of this section, we give an overview of our approach.

AI-Noether Pipeline

Our goal is to identify candidate missing polynomial axioms $A_{k+1}, \dots, A_{k+r}$ that, along with a known set of axioms $A_1, \dots, A_k$, can be used to derive a target hypothesis Q . In this study, we focus on systems defined over real polynomials with real solutions, though the approach extends naturally to complex equations (see Supplementary Note 5). The method consists of three main steps: Encode, Decompose, and Reason, illustrated in Figure 3 and summarized in Algorithm Summary.

Algorithm Summary

- 1: Input: Axioms $A_1, \dots, A_k$ and hypothesis Q
 - 2: Encode: Construct the solution set (variety) of $\{A_1, \dots, A_k, Q\}$
 - 3: Decompose: Break the variety into irreducible components
 - 4: If noisy: Use numerical methods (witness sets + symbolic regression)
 - 5: If exact: Use symbolic primary decomposition
 - 6: **for** each component **do**
 - 7: Extract candidate axioms (generators or fitted polynomials)
 - 8: Reason: Test whether candidates enable derivation of Q
 - 9: If noisy: Use theorem prover with existential quantifiers
 - 10: If exact: Use algebraic projection via Gröbner bases
 - 11: **end for**
 - 12: Output: Candidate axiom sets $\{\hat{A}_{k+1}, \dots, \hat{A}_l\}$ sufficient to explain Q
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The first step is to encode. Given a collection of known axioms $A_1, \dots, A_k$ and a hypothesis Q , we form the solution set of the system $\{A_1, \dots, A_k, Q\}$. Each equation is defined over some subset of n real variables, and so defines a hypersurface in $\mathbb{R}^n$. The intersection of these hypersurfaces defines a geometric object referred to as the variety corresponding to the system, denoted $V(A_1, \dots, A_k, Q)$. This geometric perspective is useful because algebraic derivability has a geometric interpretation: if Q is algebraically derivable from $A_1, \dots, A_k$, then Q can be written as an algebraic combination $Q = \sum_{i=1}^k \alpha_i A_i$ where each α_i is a real polynomial. Then Q vanishes whenever all the axioms vanish. So we can recast the idea of Q being derivable from any set of axioms $A_1, \dots, A_{k+r}$ as a geometric operation. Our first step in AI-Noether’s process then will be to take the known axioms $A_1, \dots, A_k$ and consequence Q and encode them into a variety.

The second step is to decompose. The variety corresponding to the equation $xy = 0$ (where x and y are real variables) is reducible as it is the union of two smaller varieties, i.e., solutions of $x = 0$ and of $y = 0$. When a variety cannot be expressed as the union of strictly smaller varieties, we call it irreducible. In writing the variety $V(xy) = V(x) \cup V(y)$ as a union of two irreducible varieties, we have decomposed it. We similarly decompose the variety $V(A_1, \dots, A_k, Q)$ into its irreducible components, i.e., a list of irreducible varieties whose union is $V(A_1, \dots, A_k, Q)$. This process is called Lasker-Noether Primary Decomposition [1], and is geometrically illustrated in Figure 4a.

If Q is not a consequence of the axioms (i.e., is not derivable from the known axioms), then $V(A_1, \dots, A_k, Q)$ contains additional irreducible components not present in $V(A_1, \dots, A_k)$. To illustrate why this matters in the context of abductive inference, assume for example that we only need one more axiom, A_{k+1} , in order to derive Q . Then $Q = \alpha_1 A_1 + \dots + \alpha_{k+1} A_{k+1}$ for unknown coefficients α_i and unknown A_{k+1} . We are of course interested then in the residual $\alpha_{k+1} A_{k+1}$. The key observation is that $V(A_1, \dots, A_k, Q)$ is the same variety as $V(A_1, \dots, A_k, \alpha_{k+1} A_{k+1})$. Similar to the example of $V(xy)$, $V(\alpha_{k+1} A_{k+1})$ is reducible. So intersecting Q with the variety corresponding to the axioms $V(A_1, \dots, A_k)$ introduces new irreducible components that are directly related to the residual $\alpha_{k+1} A_{k+1}$. Decomposition isolates these structures. So the next step in AI-Noether’s system is to take the encoded variety and decompose it into its irreducible components. The key difference from traditional symbolic regression-based discovery is that we do not assume any data for the unknown axiom or any prior knowledge of the variables being fitted.

When consequences contain noise - for instance, when Q has been inferred from experimental data - exact algebraic methods become numerically unstable. In such cases, we employ a numerical irreducible decomposition [3] via homotopy continuation, which intersects the variety with generic hyperplanes until a finite set of points called a witness set is obtained, then uses curve tracing to identify points lying on the same irreducible component. This produces a finite set of sample points on each component rather than symbolic generators. We then perform symbolic regression on these witness points to recover polynomial relations that approximately vanish on each component with an approach similar to vanishing component analysis [31]. Given a chosen monomial basis $\{M_1, \dots, M_K\}$ and N sampled witness points, the design matrix $\Phi \in \mathbb{C}^{N \times K}$ is the Vandermonde-like matrix whose entry Φ_{jk} is M_k evaluated at the j^{th} witness point; we then take as fitted coefficients the right singular vector of Φ corresponding to its smallest singular value. The smallest singular value of the resulting design matrix serves as a quality metric: low values indicate the recovered relation nearly vanishes on the component, suggesting it governs that portion of the solution space. This numerical workflow, with the numerical witness set computation and our symbolic regression step, enables decomposition even when consequences carry numerical error. This workflow is detailed in Supplementary Note 3.

The third step is to reason. For each irreducible component, we extract its defining polynomials and test whether any subset $\{\hat{A}_{k+1}, \dots, \hat{A}_l\}$, when added to the known axioms, is sufficient to derive Q . This is done by forming the variety $V(A_1, \dots, A_k, \hat{A}_{k+1}, \dots, \hat{A}_l)$ and projecting it onto the variables appearing in Q , illustrated in Figure 4b. If this projection lies within the solution set of Q , then Q is implied by the augmented system. Such sets are returned as candidate missing axioms. If no such set exists, we conclude that no single missing axiom explains Q . This is the final step of AI-Noether where we reason with each decomposed component to discover which axiom candidates can derive Q .

When axioms or consequences contain numerical constants from noisy data, algebraic projection may be overly sensitive to coefficient errors. As an alternative to algebraic derivability, for robustness to noise, we use a theorem prover to assess whether the conjecture follows logically from the augmented background theory (the original axioms together with the abduced ones). Building on AI-Descartes [9], which introduced existential derivability to handle numerical constants in data-derived conjectures, we introduce the existential constants also in the abduced axioms. The conjecture is then deemed derivable when there exists an instantiation of these abstracted constants that makes it logically implied by the background theory. This extension allows the prover to handle noisy numerical parameters and provides a principled way to test whether the augmented axioms suffice to entail the conjecture.

Using the encode, decompose, and reason steps, AI-Noether handles both exact symbolic computation and noisy numerical scenarios. For a detailed development of the algebraic and logical foundations underlying the three-step pipeline as well as all the relevant key definitions and theorems, see Supplementary Note 3. For the details of our implementation, see Supplementary Note 5. Having established the conceptual framework, we demonstrate AI-Noether’s capabilities across a range of scientific domains in the Methods, which presents experimental results organized by discovery type: recovering individual axioms (Single Missing Axiom), discovering multiple axioms simultaneously (Multiple Missing Axioms and Multiple Consequences), and handling noisy consequences (Noisy Consequences). These experiments span classical mechanics, special relativity, orbital dynamics, and electromagnetic theory, illustrating the generality of the abductive inference approach.

Discussion

Abductive Inference

This work introduces AI-Noether, an automated abductive reasoning system that connects AI-generated scientific hypotheses with established theoretical knowledge. Given a set of background axioms and a hypothesis not supported by existing theory, AI-Noether discovers a set of additional axioms that reconcile the hypothesis with the theory, provided both are expressible as polynomials. By combining algebraic-geometry tools with symbolic verification, the system provides candidate explanations and formal guarantees for deriving target hypotheses from the augmented axioms.

AI-Noether demonstrates effectiveness in establishing a breadth of known scientific laws and positions axiomatic discovery as a fundamental form of scientific inquiry. AI-Noether supports the assessment and theoretical integration of new hypotheses (e.g., those generated by data-driven or

theory-based methods) by determining whether and under what axioms they can be integrated into coherent theoretical frameworks. In this sense, the system advances the scientific method by explicitly addressing the problem of identifying and explaining the underlying principles governing the environment in which a phenomenon occurs.

We demonstrate the effectiveness of AI-Noether across a broad set of physical systems, including classical mechanics, electromagnetism, and semiconductor physics (see the Methods). We show that it robustly recovers missing axioms under both noiseless and noisy conditions. Comparisons against solver-based methods and language model baselines highlight our approach’s advantages in correctness and formal verifiability. More broadly, applying AI-Noether to domains where current theories are incomplete or internally inconsistent points toward a shift in the scientific method itself: one in which abductive inference both explains existing discoveries better and uncovers new principles that reveal the underlying structure of physical theories.

Scope, Recoverability and Validation

AI-Noether is a purely algebraic system for generating a set of candidate missing axioms from a polynomial hypothesis and a background theory expressible as polynomial equations. We employ the same techniques as in AI-Hilbert to deal with non-algebraic functions, derivatives, and integrals. The first method is that we simplify integrals and derivatives as much as possible and then assign new variables to any remaining integrals, derivatives, and non-algebraic functions. The second is that we treat derivatives symbolically, treating them as indeterminates in a polynomial. For trigonometric functions (as in the case of the simple harmonic oscillator, light damping, and Compton Scattering benchmarks in the Methods), we can consider trigonometric polynomials by looking at the corresponding axioms as polynomials in the respective trigonometric functions. This becomes more challenging, however, when dealing with trigonometric functions composed with other polynomial functions. It will be interesting therefore to extend AI-Noether to handle cases when these techniques are not sufficient to derive the consequences from the full set of axioms.

Naturally, when the dimensions lost by removing axioms cannot be recovered by adding consequences, no irreducible component of the encoded variety will contain the missing axioms, and AI-Noether returns an empty candidate set. This behavior follows from the geometric reading of Theorem 1 in the Criteria for Recoverability of Axioms subsection of the Methods: each axiom defines a hypersurface that reduces the dimension of the solution space by at most one, and each consequence introduces a new cut (a new hypersurface constraint), so when the cuts contributed by the available consequences are insufficient to compensate for the removed axioms, the decomposition cannot isolate a component that carries the missing information. We illustrate this concretely in Scenario 4 of Supplementary Note 4, which examines the exact output of AI-Noether in the context of the results in the Multiple Missing Axioms subsection of the Methods: removing both the gravitational law and the center of mass equation $m_1 d_1 = m_2 d_2$ from Kepler’s third law yields Saved combos: 0 and Strong candidates: 0 and not some spurious candidate set, reflecting that the system reports failure when a recovery cannot be justified by the available theory and consequences.

Moreover, AI-Noether can recover a missing axiom only if each variable appearing in that axiom also occurs in at least one of the available axioms or in the target hypothesis. If a missing axiom involves latent variables that are entirely absent from the observed theory and consequences, the current framework cannot identify or introduce them automatically. However, as shown in the Multiple Missing Axioms subsection of the Methods with Kepler’s third law and multiple missing axioms, AI-Noether can, in some cases, recover information about variables involved in multiple missing axioms by discovering coupled versions of the missing axioms and variables. When recovery is not possible because the missing axioms involve latent variables, AI-Noether produces the same empty candidate set shown in Scenario 4 of Supplementary Note 4, rather than fabricating a candidate that omits the latent variables.

A related practical limitation is the choice of background axioms supplied to AI-Noether. In principle, scientific domains may contain many plausibly relevant axioms, but including all of them is often computationally intractable. Thus, as in automated theorem proving literature [51], AI-Noether may require a premise-selection step: selecting a relevant subset of background knowledge axioms before attempting abduction. This is analogous to premise selection in theorem proving [52], the task of identifying the axioms most likely to be useful for proving a target statement. In the experiments presented here, the background theories are manually curated to isolate the abductive task, but

in larger theories this selection could be performed by domain experts, learned classifiers, retrieval systems, LLM-based tools, or other hypothesis- and premise-selection methods [54].

A further limitation concerns the validation of genuinely novel hypotheses. In the historical examples considered in our work, the inferred axioms can be compared with known ground-truth principles. In genuinely novel settings, however, the physically correct axioms are unknown. AI-Noether can therefore provide internal validation only: it can check whether or not the extended axiom system makes the target hypothesis derivable and whether or not the added axioms introduce algebraic inconsistencies. Establishing physical correctness still requires external validation, such as experiments, future observations, or expert assessment. If the starting background theory is itself inconsistent, the abductive problem also becomes logically ill-posed, since inconsistent theories may make arbitrary hypotheses derivable.

Finally, although effective on a wide range of benchmark systems, AI-Noether relies on algebraic decomposition and symbolic verification whose computational cost can grow rapidly with the number of variables, axioms, and polynomial degrees, depending on the system.

Improving the Generality of AI-Noether

In this study, we have focused on developing what we believe is a first-of-its-kind abductive inference system and testing it on existing laws of nature. A natural next step is therefore to employ AI-Noether in scientific domains where existing theories are incomplete or internally inconsistent, with the aim of supporting genuinely novel discoveries and of discovering new axioms as substantive scientific contributions that extend or revise existing theories. We also aim to extend AI-Noether beyond equations over a polynomial basis to handle differential operators and inequality constraints, enabling abductive inference across a broader range of scientific domains.

Methods

In this section, we present the results from testing our system on various real-world physical theories. We test AI-Noether on 9 systems reported in AI-Hilbert [11] as well as 3 additional systems of varying complexity, the most complex being the results from a recent work on the Carrier-Resolved Photo-Hall Effect [23]. The other two systems include a combined axiomatic system involving Einstein’s relativistic mass, length contraction, and time dilation laws as consequences, and finally a baseline case of a simple harmonic oscillator. We illustrate how AI-Noether performs under different settings using different examples.

Single Missing Axiom

Example: Carrier-Resolved Photo-Hall Effect

The Carrier-Resolved Photo-Hall effect [23] equation describes the relationship between various parameters of a semiconducting surface (illustrated in Supplementary Fig. 5) and is given by:

$$Q := H - \frac{r e \mu_P^2 [p_0 + \Delta n(1 - \beta^2)]}{\sigma^2} = 0, \quad (1)$$

where $\sigma = e\mu_P[p_0 + \Delta n(1 + \beta)]$ is the conductivity. We assume the axioms

$$A_1 : \beta\mu_P - \mu_N \quad \text{Mobility ratio definition} \quad (2)$$

$$A_2 : \mu_H - r\mu \quad \text{Hall factor correction} \quad (3)$$

$$A_3 : p_h - p_0 - \Delta p \quad \text{Hole concentration balance} \quad (4)$$

$$A_4 : n - \Delta n \quad \text{Electron concentration tracking} \quad (5)$$

$$A_5 : \Delta p - \Delta n \quad \text{Charge neutrality condition} \quad (6)$$

$$A_6 : \sigma - e p_h \mu_P - e n \mu_N \quad \text{Total conductivity sum} \quad (7)$$

$$A_7 : H(p_h + \beta n)^2 e - r p_h + r \beta^2 n \quad \text{Hall coefficient relation.} \quad (8)$$

Here μ_N and μ_P are electron and hole drift mobilities, $\beta = \mu_N/\mu_P$ is the mobility ratio (encoded by A_1), μ_H is the Hall mobility with Hall scattering factor r and mobility μ related by $r = \mu_H/\mu$ (encoded by A_2), Δn and Δp are the electron and hole photo-carrier densities (equal in steady state by A_5), H is the Hall coefficient, σ is the conductivity, n is the electron density, p_0 is the background hole density, and p_h is the hole density (for a p-type material).

The photo-Hall effect measurement begins with the semiconductor in its dark equilibrium state, where only majority carriers (e.g., holes with concentration p_0 in p-type material) contribute to transport, establishing baseline conductivity σ_0 and Hall coefficient H_0 . Upon illumination with above-bandgap photons, electron-hole pairs are generated satisfying charge neutrality $\Delta n = \Delta p$, increasing the total hole concentration to $p_h = p_0 + \Delta p$ and electron concentration to $n \approx \Delta n$.

An alternating current (AC) magnetic field from the rotating parallel dipole line (PDL) system deflects both carrier types via the Lorentz force, generating a Hall voltage across the sample. As light intensity I varies, both conductivity $\sigma(I) = e(p_h\mu_P + n\mu_N)$ and Hall coefficient $H(I)$ evolve non-linearly due to competing contributions from holes and electrons with different mobilities. The key innovation is the relation $\Delta\mu_H = \frac{d(\sigma^2 H)}{d\sigma}$, which extracts the hole-electron mobility difference from the measured transport data.

An exact derivation of Q from the axioms, both directly and as an algebraic combination, can be found in Supplementary Note 4. However, it suffices to recognize that, because Q is algebraically derivable from the known axioms, we can write

$$Q = \sum_{i=1}^7 \alpha_i A_i \quad (9)$$

where the α_i are polynomials (Supplementary Note 4) and $i \in \{1, \dots, 7\}$.

One of the key axioms in the derivation is axiom A_7 , which relates the Hall coefficient to all the essential quantities. In our setting, we assume that we neither have A_7 nor any of the multipliers $\alpha_1, \dots, \alpha_7$. Using methods described in [23, 48, 9, 11], we can recover Q from data. Encoding the known information $A_1, \dots, A_6, Q$ into a variety, decomposing it, and passing it through the reasoning step yields the following polynomials.

Decomposition:

$$\text{Variety 1: } \langle \sigma, \mu_P, n - \Delta n, \mu_N, \Delta p - \Delta n, p_h - p_0 - \Delta n, \mu r - \mu_H \rangle \quad (10)$$

$$\text{Variety 2: } \langle \sigma, e, n - \Delta n, \Delta p - \Delta n, p_h - p_0 - \Delta n, \mu_N - \mu_P\beta, \mu r - \mu_H \rangle \quad (11)$$

$$\begin{aligned} \text{Variety 3: } & \langle n - \Delta n, \Delta p - \Delta n, p_h - p_0 - \Delta n, \mu_N - \mu_P\beta, \mu r - \mu_H, \\ & \mu_N e \Delta n + e \mu_P p_0 + e \mu_P \Delta n - \sigma, e \mu_P p_0 + e \mu_P \Delta n \beta + e \mu_P \Delta n - \sigma, \\ & \sigma H(p_0 + \Delta n(1 + \beta)) + r(\mu_N \beta \Delta n - \mu_P(p_0 + \Delta n)), \\ & \mu_N^2 \Delta n r + \mu_N \Delta n \sigma H - \mu_P^2 p_0 r - \mu_P^2 \Delta n r + \mu_P p_0 \sigma H + \mu_P \Delta n \sigma H, \\ & r \mu_P(p_0 + \Delta n(1 - \beta^2)) - \sigma H(p_0 + \Delta n(1 + \beta)), \\ & - e \sigma H(p_0 + \Delta n(1 + \beta)) + r(\sigma(1 - \beta) + e \mu_P \beta p_0), \\ & \mu_N e \mu_P p_0 r - \mu_P \beta r \sigma + \mu_P r \sigma - \sigma^2 H, \mu_N^2 e p_0 r - \mu_P \beta^2 r \sigma + \mu_P \beta r \sigma - \beta \sigma^2 H, \\ & \mu \sigma^2 H - \mu_N \mu_H e \mu_P p_0 + \mu_H \mu_P \beta \sigma - \mu_H \mu_P \sigma, \mu \beta \sigma^2 H - \mu_N^2 \mu_H e p_0 + \mu_H \mu_P \beta^2 \sigma - \mu_H \mu_P \beta \sigma, \\ & \mu p_0 \sigma H + \mu \Delta n \beta \sigma H + \mu \Delta n \sigma H - \mu_H \mu_P p_0 + \mu_H \mu_P \Delta n \beta^2 - \mu_H \mu_P \Delta n, \\ & \mu \mu_N \Delta n \sigma H + \mu \mu_P p_0 \sigma H + \mu \mu_P \Delta n \sigma H + \mu_N \mu_H \mu_P \Delta n \beta - \mu_H \mu_P^2 p_0 - \mu_H \mu_P^2 \Delta n, \\ & e H(p_0 + \Delta n(1 + \beta))^2 - r(p_0 + \Delta n(1 - \beta^2)) \rangle \quad (12) \end{aligned}$$

Generators selected by Reasoning module:

$$\sigma H(p_0 + \Delta n(1 + \beta)) + r(\mu_N \beta \Delta n - \mu_P(p_0 + \Delta n)) \quad (13)$$

$$r \mu_P(p_0 + \Delta n(1 - \beta^2)) - \sigma H(p_0 + \Delta n(1 + \beta)) \quad (14)$$

$$- e \sigma H(p_0 + \Delta n(1 + \beta)) + r(\sigma(1 - \beta) + e \mu_P \beta p_0) \quad (15)$$

$$e H(p_0 + \Delta n(1 + \beta))^2 - r(p_0 + \Delta n(1 - \beta^2)) \quad (16)$$

Notice that each of the obtained relationships is the exact Hall coefficient relation with other axiom substitutions applied. The first equation is A_7 with A_3, A_4, A_5 substituted, multiplied by μ_P , and

rewritten using A_1 and A_6 . The second is the negative of the first, with A_1 substituted in as well. The third is the same as the second, except that A_6 is substituted and it is multiplied by e . The last is A_7 with only A_3, A_4, A_5 substituted.

Each of these axioms contains the same information as A_7 . The alternate expressions reflect the fact that the expression of A_7 is not unique. We could have defined the same relation in multiple forms.

Additionally, if we start with an overly strong assumption, such as $\sigma = 0$ or $\mu_P = 0$ from Variety 1, we could also derive $Q = 0$. We provide an additional layer of filtering for such cases when using projections to ensure that we do not eliminate more variables than necessary. A discussion of this can be found in Supplementary Note 5.

Example: Simple Harmonic Oscillator

For a simpler and more illustrative example, consider the case of a simple harmonic oscillator with a point mass m and length L , with a force F acting on it creating a small angle θ , illustrated in Supplementary Fig. 6. Taking g as the acceleration constant, at j iterations, we have the equation that describes the time period T_j given by:

$$Q := T_j - 2\pi j \sqrt{\frac{L}{g}} \quad (17)$$

whose polynomial form, $gT_j^2 - 4\pi^2 j^2 L = 0$, can be derived (either traditionally or as an algebraic combination) from the following axioms:

$$A_1 : a_d - g \sin \theta \quad \text{Gravitational component of acceleration} \quad (18)$$

$$A_2 : d - L\theta \quad \text{Height of oscillator} \quad (19)$$

$$A_3 : T\omega - 2\pi \quad \text{Frequency-period relation} \quad (20)$$

$$A_4 : d\omega^2 - a_d \quad \text{Acceleration equation} \quad (21)$$

$$A_5 : T_j - jT \quad \text{Time period over } j \text{ iterations} \quad (22)$$

$$A_6 : \sin \theta - \theta \quad \text{Small angle assumption} \quad (23)$$

where a_d is the acceleration of the object, ω is the frequency, and T is the time period for one oscillation. A small detail here is that in the algebraic context, like in [11], we treat $\sin \theta$ and π symbolically. One of the key axioms here is A_1 , the gravitational component of acceleration. If we remove this assumption from our system, then we can still recover Q from data, as seen in [48]. Applying AI-Noether to Q and the known axioms results in the following output:

Decomposition:

$$\text{Variety 1: } \langle \theta - \sin \theta, 2j\pi - \omega T_j, Tj - T_j, L \sin \theta - d, dg - a_d L, g \sin \theta - a_d, \\ T\omega - 2\pi, 2\omega L\pi - Tg, 2\omega d\pi - a_d T, \omega^2 L - g, T^2 g - 4L\pi^2, \omega^2 d - a_d, a_d T^2 - 4d\pi^2 \rangle \quad (24)$$

Generators selected by Reasoning module:

$$g \sin \theta - a_d \quad (25)$$

Results Summary on Other Problems

We report the performance of AI-Noether on each test system in Table 1. In each case, we iterate through the axioms, omit an axiom from the system, and test if we can recover the missing axiom while not making any assumption about it. In almost every case, we were able to recover the missing axiom. Supplementary Note 4 provides a further discussion of some of these systems. We will discuss the problem of the relativistic laws in the Multiple Consequences subsection, where we only partially recover missing axioms correctly while recovering alternate axioms for some cases.

Multiple Missing Axioms

We now examine the problem of recovering more than one axiom to explain a hypothesis Q . In this case, we have

$$Q = \sum_{i=1}^{k+r} \alpha_i A_i \quad (26)$$

However, we only have access to $A_1, \dots, A_k$. We seek a candidate set of axioms that, together with $A_1, \dots, A_k$, suffices to derive Q .

Example: Kepler's Third Law

Consider Kepler's third law, which relates the orbital period of two celestial bodies to their distances and masses. It can be written as

$$Q := p - \sqrt{\frac{(d_1 + d_2)^3}{G(m_1 + m_2)}} = 0, \quad (27)$$

where p is the orbital period (scaled by a factor of 2π), m_1, m_2 the masses, and d_1, d_2 their distances from the center of mass. As in [11], we will scale units to have $G = 1$ for consistency with their framework, though this is not necessary. For circular orbits, we use the following axioms (with p non-dimensionalized so that $\omega p = 1$), illustrated in Supplementary Fig. 7:

$$A_1 : (d_1 + d_2)^2 F_g - m_1 m_2 \quad \text{Gravitational Force} \quad (28)$$

$$A_2 : F_c - m_2 d_2 \omega^2 \quad \text{Centrifugal Force} \quad (29)$$

$$A_3 : F_g - F_c \quad \text{Newton's third law} \quad (30)$$

$$A_4 : \omega p - 1 \quad \text{Frequency-period relation} \quad (31)$$

$$A_5 : m_1 d_1 - m_2 d_2 \quad \text{Center of mass equation} \quad (32)$$

to derive

$$(m_1 + m_2)p^2 - (d_1 + d_2)^3 = 0, \quad (33)$$

which is Kepler's law in polynomial form.

Assume that we are missing axioms A_3 and A_4 - the relationship between centrifugal and gravitational force and the relationship between frequency and period. Then, encoding the known information - equations A_1, A_2, A_5 , and Q - our system returns the following relevant component of the primary decomposition and selected generator from the reasoning module:

Selected varieties from Decomposition module:

$$\begin{aligned} \text{Variety 1: } \langle & m_1 d_1 - m_2 d_2, F_g p^2 - m_2 d_2, F_g(d_1^2 + 2d_1 d_2 + d_2^2) - m_1 m_2, \\ & d_1^3 + 2d_1^2 d_2 + d_1 d_2^2 - m_2 p^2, m_1 p^2 - d_1^2 d_2 - 2d_1 d_2^2 - d_2^3, \\ & F_c - \omega^2 m_2 d_2, F_c p^2 - \omega^2 d_1^3 d_2 - 2\omega^2 d_1^2 d_2^2 - \omega^2 d_1 d_2^3 \rangle \end{aligned} \quad (34)$$

Generators selected by Reasoning module:

$$F_g p^2 - m_2 d_2 \quad (35)$$

We return $F_g p^2 - m_2 d_2$. This equation is the same as the centrifugal force equation A_2 but with the missing relationships $F_c = F_g$ and $\omega = \frac{1}{p}$ substituted in. Therefore, the missing information is discovered, since equation A_2 is assumed to be known, although it is not decoupled. To go one step further, we can remove three out of the five axioms: equations A_2, A_3 , and A_4 , leaving us with only the gravitational force equation A_1 and the center of mass equation A_5 . Similarly, running our system with the Kepler hypothesis gives us a similar component:

Selected varieties from Decomposition module:

$$\begin{aligned} \text{Variety 1: } \langle & m_1 d_1 - m_2 d_2, F_g p^2 - m_2 d_2, F_g(d_1^2 + 2d_1 d_2 + d_2^2) - m_1 m_2, \\ & d_1^3 + 2d_1^2 d_2 + d_1 d_2^2 - m_2 p^2, m_1 p^2 - d_1^2 d_2 - 2d_1 d_2^2 - d_2^3 \rangle \end{aligned} \quad (36)$$

Generators selected by Reasoning module:

$$F_g p^2 - m_2 d_2 \quad (37)$$

We get the same axiom candidate. However, whereas in the previous case we knew the centrifugal force equation, we now assume that we do not. So we are not just recovering the relationships $F_c = F_g$ and $\omega p = 1$, we are also recovering the centrifugal force axiom itself $F_c = m_2 d_2 \omega^2$ with the former relations coupled with it. A criterion for exactly recovering axioms is given in the Criteria for Recoverability of Axioms subsection.

Table 2 shows the results of attempting to recover various combinations of pairs and triples of axioms for Kepler’s third law.

Results Summary on Other Problems

Across twelve benchmark systems, we systematically removed axiom pairs and triples and attempted to recover explanations for Q and have summarized our results in Table 3. Success rates vary considerably by problem complexity and number of missing axioms. For 2-axiom recovery, success rates range from 10% (Time Dilation, Light Damping) to 81% (Relativistic Laws), with a median of 56%. For 3-axiom recovery, performance degrades substantially: success rates drop to 0-69%, with most systems achieving 10-25% recovery. The best-performing systems are Relativistic Laws (81% pairs, 69% triples), Escape Velocity (80% pairs, 40% triples), Kepler (70% pairs, 20% triples), and Simple Harmonic Oscillator (67% pairs, 20% triples). The Hall Effect and Compton Scattering systems, despite their larger axiom sets (7 and 10 axioms), achieve moderate success (62% and 38% for pairs; 49% and 21% for triples, respectively). The Carrier-Resolved Photo-Hall system (62% pairs, 43% triples) demonstrates the method’s scalability to complex systems with 7 axioms and 15 variables. Overall, recovery difficulty increases with the number of missing axioms and, to a lesser extent, with system size, though the relationship is non-monotonic.

Clearly having more information is helpful, and so the success rate is higher in cases where we were only missing two axioms rather than three. The highest success rate was for the Relativistic Laws (81% pairs, 69% triples), likely because this is the only system with not one but three consequences that we are drawing information from, as we will see in the Multiple Consequences subsection.

Multiple Consequences

As we will see in the Criteria for Recoverability of Axioms subsection, having multiple consequences can help us recover decoupled axioms in the case where multiple axioms are missing. We demonstrate AI-Noether’s ability to handle multiple related consequences simultaneously, exploring relativistic phenomena. Here, each consequence Q_i is the sum

$$Q_i = \sum_{j=1}^{k+r} \alpha_{i,j} A_j \quad (38)$$

where each $\alpha_{i,j}$ is unknown and $A_{k+1}, \dots, A_{k+r}$ are unknown as well.

Example: Relativistic Laws

Consider Einstein’s relativistic mass, length contraction, and time dilation formulae, which describe how mass increases and lengths contract for objects moving at relativistic speeds, illustrated in Supplementary Fig. 8. These can be expressed in polynomial form as:

$$Q_1 := -c^2 f^2 + f_0^2 (c^2 - v^2) = 0 \quad \text{Time dilation} \quad (39)$$

$$Q_2 := -m_0^2 c^2 + m^2 (c^2 - v^2) = 0 \quad \text{Relativistic mass} \quad (40)$$

$$Q_3 := L^2 c^2 - L_0^2 (c^2 - v^2) = 0 \quad \text{Length contraction} \quad (41)$$

These consequences arise from a unified set of axioms describing light clock geometry in different reference frames. The axiom structure, illustrated in Figure 5, reveals how different subsets of axioms are required to derive each consequence: axioms A_1 - A_4 form the foundation for time dilation (Q_1), with additional axioms A_5 - A_6 needed for relativistic mass (Q_2), and additional axiom A_7 required for length contraction (Q_3).

$$A_1 : f_0 t_0 - 1 \quad \text{Frequency-period (rest)} \quad (42)$$

$$A_2 : ft - 1 \quad \text{Frequency-period (moving)} \quad (43)$$

$$A_3 : ct_0 - 2L_0 \quad \text{Perpendicular light travel (rest)} \quad (44)$$

$$A_4 : \frac{c^2 t^2}{4} - L_0^2 - \frac{v^2 t^2}{4} \quad \text{Light path geometry (moving)} \quad (45)$$

$$A_5 : m_0 u_0 - m u_y \quad \text{Transverse momentum conservation} \quad (46)$$

$$A_6 : u_0 t_0 - u_y t \quad \text{Transverse distance invariance} \quad (47)$$

$$A_7 : t(c^2 - v^2) - 2Lc \quad \text{Parallel light travel time} \quad (48)$$

Axioms A_1 and A_2 define the period-frequency relationship in the rest and moving frames, respectively. Axiom A_3 encodes that light travels distance $2L_0$ in time t_0 for a perpendicular round trip in the rest frame. Axiom A_4 captures the geometric constraint from a perpendicular light clock: a moving observer sees light travel along the hypotenuse of a right triangle, relating the rest-frame quantities to the moving-frame time and velocity. Axioms A_5 and A_6 ensure conservation of momentum and distance in the transverse (perpendicular) direction. Finally, axiom A_7 describes light traveling parallel to the direction of motion, accounting for the asymmetric forward and backward travel times.

The derivation structure proceeds as follows. From A_3 and A_4 , we obtain $t^2(c^2 - v^2) = c^2 t_0^2$, the basic time dilation relation. Substituting the frequency-period relations from A_1 and A_2 yields Q_1 . For relativistic mass (Q_2), we use A_6 to relate transverse velocities through the time dilation factor, then apply momentum conservation from A_5 to obtain the mass transformation. For length contraction (Q_3), axiom A_7 provides the parallel light travel constraint, which when combined with the time dilation relation and A_3 gives the length contraction formula.

We test AI-Noether's ability to recover missing axioms in two scenarios: single-axiom removal and multi-axiom removal with multiple consequences.

We first consider the single-axiom case. We saw in Table 1 that the combined relativistic laws were the only case where there were two axioms which we do not recover exactly. When removing individual axioms and attempting recovery using a single consequence, AI-Noether exhibits interesting behavior for axioms A_4 and A_7 . Instead of recovering the original quadratic forms, the system returns factorized alternatives. For instance, when A_4 is removed, AI-Noether recovers the pair:

$$c - 2f_0^2 Lt \quad \text{and} \quad c + 2f_0^2 Lt. \quad (49)$$

Neither factor reproduces A_4 . Rather, the first factor, together with the two surviving moving-frame axioms $A_2 : ft - 1$ and $A_7 : t(c^2 - v^2) - 2Lc$, is algebraically equivalent to the target consequence Q_1 itself. Indeed, one has the polynomial identity

$$t^2 Q_1 = -c(c - 2f_0^2 Lt) - c^2(ft + 1)A_2 + f_0^2 t A_7, \quad (50)$$

so that, modulo $A_2 = 0$ and $A_7 = 0$ and for $c, t \neq 0$, the relation $c - 2f_0^2 Lt = 0$ holds exactly when $Q_1 = 0$. The system thus closes the derivability gap by re-expressing the consequence through the remaining axioms rather than by recovering the missing geometric constraint A_4 , which is why A_4 is reported as not recovered in Table 1. Note that A_4 depends on v whereas the recovered linear factors do not: the missing v -dependence is supplied entirely by A_7 . A similar factorization occurs for A_7 . The main algebraic structure here is that the solution variety of the quadratic constraint decomposes into a union of two linear varieties (hyperplanes), each corresponding to one factor.

What we see through the primary decomposition is that when A_4 is removed, the variety $V(\{A_1, A_3, A_5, A_6, A_7, Q\})$ decomposes into components whose generators include these linear factors. These factors do not individually reproduce the physics of A_4 , but they appear in the decomposition because they define subvarieties that, when intersected with the other axioms, collectively constrain the solution space in a way that allows Q to be derived. This shows that abductive inference may return algebraic expressions that are structurally different from the original axioms yet sufficient to derive the target consequence when combined with background knowledge. The system identifies what is algebraically necessary to close the gap between axioms and consequence, which may differ from the canonical physical formulation of the missing law.

In the multi-axiom case with multiple consequences, we remove three axioms simultaneously: A_2 (required for Q_1), A_3 (required for Q_1 and Q_3), and A_5 (required for Q_2). Running AI-Noether with all three consequences Q_1, Q_2, Q_3 and the remaining axioms yields:

Generators selected by Reasoning module:

$$ft - 1 \quad (51)$$

$$ct_0 - 2L_0 \quad (52)$$

$$m_0u_0 - mu_y \quad (53)$$

Notably, the system recovers all three missing axioms exactly – A_2, A_3 , and A_5 – without the coupling artifacts observed in the Multiple Missing Axioms subsection. So, having multiple consequences has an important advantage: the additional constraints allow the decomposition to isolate the individual missing axioms rather than returning their coupled combinations. As we formalize in the Criteria for Recoverability of Axioms subsection, multiple consequences increase the algebraic constraints on the variety, forcing the primary decomposition to separate axioms that would otherwise appear only in a coupled form.

Noisy Consequences

We now consider the effect of noise in the consequences. For each of the problems in this section, as well as with the remaining test problems, we ran AI-Hilbert [11] on synthetically generated datasets for Q (respectively, Q_i for relativistic laws) with standard Gaussian noise added to the data proportional to the means of each column with a noise level ϵ . For each variable in the system, we use AI-Hilbert’s synthetic data generator to generate a column of 1000 data points for each independent variable and use the ground-truth equation to generate data for the dependent variable. To each column, we then add independent standard Gaussian noise with the standard deviation set to ϵ times the mean of that column. We then run AI-Hilbert on the generated data as well as known axioms and use the output of AI-Hilbert as the hypothesis to be explained. For further details about the data-generation mechanism, see [11]. We therefore obtain perturbations of each consequence rather than the true consequences themselves. We illustrate this with the following two examples of Kepler’s third law and relativistic laws continued from earlier.

Example with Single Axiom Correction: Kepler, Continued

Taking the previous example of Kepler’s third law, assume that the gravitational force axiom $F_g(d_1 + d_2)^2 - m_1m_2$ is missing. With a noise level of $\epsilon = 1\%$, the resulting consequence Q that was obtained is given by:

$$-1.01387623m_2d_2^3 - 2.02927363m_2d_1d_2^2 - 1.01837465m_2d_1^2d_2 + 1.0142876m_1m_2p^2 = 0 \quad (54)$$

This, however, does not come with a certificate of derivability, since the remaining axioms cannot derive this consequence. We assume as before that we have no data for the missing axiom, so we cannot use a traditional symbolic regression approach to determine the missing axiom. Algebraically computing a primary decomposition results in obtaining Q as the only axiom candidate. Since algebraic decomposition uses exact computation, we can only handle exact derivability via algebraic projection. Therefore, we turn to a numerical irreducible decomposition that attempts to generate approximate points on the irreducible components, and then perform symbolic regression on these points. The data generated is defined over all the variables of the system, not just those in the consequence or any single axiom; therefore, there is a hyperparameter of which variables to fit to that is to be tuned. Of course, one could choose various hyperparameters to get a family of axiom candidates. Fitting the data generated in our system yields the following axiom candidate, which we then verify using our reasoning module.

Axiom generated by Decomposition and Reasoning modules:

$$-0.502463F_gd_1^2 - 1.012319F_gd_1d_2 - 0.497537F_gd_2^2 + 0.497536m_1m_2 \quad (55)$$

Notice that this indeed is a noisy perturbation of the missing axiom: $F_g(d_1^2 + 2d_1d_2 + d_2^2) - m_1m_2$.

Example: Relativistic laws with noisy consequences

We test the robustness of our approach by combining multiple challenges: noisy consequences, multiple missing axioms, and multiple consequences. Running AI-Hilbert [11] to generate noisy

versions of the three relativistic consequences from synthetically generated data with 1% Gaussian noise (standard deviation proportional to the mean of each column) gives the following perturbations of the relativistic laws (rounded to four decimal places for brevity):

$$\tilde{Q}_1 := -1.0089c^2f^2 + 0.9986f_0^2c^2 - 0.9956f_0^2v^2 = 0 \quad (\text{Time dilation}) \quad (56)$$

$$\tilde{Q}_2 := -0.9876m_0^2c^2 + 1.0134m^2c^2 - 1.0089m^2v^2 = 0 \quad (\text{Relativistic mass}) \quad (57)$$

$$\tilde{Q}_3 := 0.9945L^2c^2 - 1.0123L_0^2c^2 + 1.0067L_0^2v^2 = 0 \quad (\text{Length contraction}) \quad (58)$$

We removed three axioms that span different roles in the derivation structure (recall Figure 5): A_2 ($ft - 1$), required for time dilation; A_5 ($m_0u_0 - mu_y$), required for relativistic mass; and A_4 ($\frac{c^2t^2}{4} - L_0^2 - \frac{v^2t^2}{4}$), the geometric constraint from the perpendicular light clock. Without these, we again do not get any certificate of derivability of the laws.

This selection is also particularly interesting because, as we saw in the Multiple Consequences subsection, axiom A_4 exhibits factorization behavior in the noiseless single-axiom case: the algebraic decomposition returns linear factors rather than the original quadratic form. This raises the question: can numerical methods recover what algebraic methods transform?

Using numerical irreducible decomposition (Supplementary Note 3) followed by symbolic regression on witness points sampled from each component, our system recovers noisy approximations of all three missing axioms:

Axioms recovered via numerical decomposition and automated reasoning:

$$1.0000 ft - 0.6599 \quad (59)$$

$$- 0.9899 m_0u_0 + 1.0000 mu_y \quad (60)$$

$$- 0.2525 c^2t^2 + 0.2451 v^2t^2 + 1.0000 L_0^2 \quad (61)$$

The recovered axioms closely approximate the true missing relationships despite 1% noise in the consequences. Most notably, axiom A_4 is recovered in its original quadratic form with coefficient errors of approximately 1-2%, rather than the factorized linear expressions produced by algebraic decomposition in the noiseless case. We see a fundamental difference between algebraic and numerical abductive inference:

- Algebraic decomposition operates purely symbolically, decomposing varieties into unions of irreducible components. When A_4 is missing, the variety structure naturally factors into linear subvarieties, yielding generators like $c \pm 2f_0^2Lt$. While algebraically sufficient for derivability, these do not match the canonical physical form.
- Numerical decomposition samples witness points on each irreducible component, then performs symbolic regression to fit polynomials that approximately vanish on those points. Because we regress using the monomial support of the original missing axiom as a basis (here, $\{c^2t^2, v^2t^2, L_0^2, 1\}$), the method directly recovers a noisy version of A_4 rather than its factorization. The witness points constrain all monomials simultaneously, bypassing the algebraic factorization that occurs in the symbolic setting.

The numerical approach has a practical advantage. By operating on sampled data rather than symbolic manipulation, it can recover axioms in forms that more closely match their canonical physical formulations. The choice of monomial basis serves as a hard constraint that forces a polynomial fit to the basis, while preserving the algebraic foundation of the decomposition step. In practice, this monomial basis is a hyperparameter that one would tune to the problem of study using domain knowledge. The complete numerical workflow - witness set computation via homotopy continuation, robust symbolic regression with outlier removal, and coefficient normalization - is detailed in Supplementary Note 3.

Results Summary on Other Problems

We summarize our results in Figure 6. For each problem, we apply the numerical irreducible decomposition (details in Supplementary Note 3) to obtain witness points on each component of the variety defined by the remaining axioms and target. We then perform symbolic regression to recover the missing axiom by fitting a polynomial in the null space of the Vandermonde-like matrix

constructed from these points. To improve robustness against numerical artifacts and outliers introduced by the homotopy continuation, we employ iterative outlier removal: points with the largest residuals (top 10% by default) are discarded, and the regression is repeated until convergence. The recovered polynomial is normalized so that its largest coefficient has magnitude one, and we apply the same normalization to the ground-truth axiom. The coefficient distance reported is the ℓ_2 norm between these normalized coefficient vectors, following [11]; since $p = 0$ and $-p = 0$ define the same constraint, we report the minimum of $\|\mathbf{c}_{\text{fit}} - \mathbf{c}_{\text{true}}\|$ and $\|\mathbf{c}_{\text{fit}} + \mathbf{c}_{\text{true}}\|$.

Failure cases in Figure 6 arise from two sources. (i) Timeouts during witness set computation or symbolic regression, which we cap at two hours by default; this time constraint primarily affected the Compton scattering results and is detailed in Supplementary Note 5. For this reason we omit Compton Scattering from Figure 6: too few of its noisy runs complete within the timeout (see Supplementary Table 4) for its curve to be meaningful, and we discuss this further in Supplementary Note 5. (ii) Cases where the reasoning module (Supplementary Note 3) cannot verify that the recovered axiom, combined with the remaining axioms, implies the target consequence. We give a discussion on the criteria for when an axiom is recoverable in the noiseless and noisy case in the Criteria for Recoverability of Axioms subsection. A natural follow-up question is regarding why some curves in Figure 6 are non-monotone in ε , i.e., regarding why the recovered coefficient error sometimes increases as the input noise decreases. The noise applied here enters the data, propagates through AI-Hilbert to produce the consequence we hand to AI-Noether, and then propagates through AI-Noether’s numerical decomposition and symbolic regression step to produce the recovered axiom. The noise that AI-Noether actually receives in the consequence is therefore the noise filtered through AI-Hilbert, and this filtering is itself not monotone in ε because AI-Hilbert’s polynomial fit is sensitive to the conditioning of the underlying problem at each noise level. To isolate AI-Noether’s contribution from this upstream effect, we report a more detailed decomposition of the noise sources in Supplementary Note 5.

Criteria for Recoverability of Axioms

Recovering axioms without noise in the consequences

The theorem below provides a sufficient criterion for recovering missing axioms. In order to state the technical details, we will use terminology from algebraic geometry. A detailed overview of all terminology relevant to the core method can be found in Supplementary Note 3.

As shown in the Primary Decomposition theorem in Supplementary Note 3, any ideal can be decomposed as an intersection of ideals whose radicals are prime. The minimal elements among these primes are called the minimal associated primes of the ideal. We will also use the notion of the height of an ideal. For a prime ideal P , we define its height, $\text{ht}(P)$, as the length n of the longest chain $P_0 \subsetneq P_1 \subsetneq \dots \subsetneq P_n = P$ of prime ideals properly contained in P . For a general ideal, its height is the minimum of the heights of its associated primes, which is the same as the minimum of the heights of all primes that contain the ideal.

Theorem 1 (Common minimal associated prime). Given ideals $I \subset J \subsetneq \mathbb{R}[x_1, \dots, x_n]$ such that $\text{ht}(I) = \text{ht}(J)$, then I and J have a common minimal associated prime.

Proof of Theorem 1. Let P be an associated prime for J such that $\text{ht}(P) = \text{ht}(J)$. Then we have that $I \subset J \subset P$. Since $\text{ht}(I) = \text{ht}(P)$, there is no prime Q such that $I \subset Q \subsetneq P$. Otherwise $\text{ht}(I) \leq \text{ht}(Q) < \text{ht}(P)$. But then P is a minimal prime containing I and thus a minimal associated prime for I .

Example: Let J be the ideal of the full set of axioms $\langle A_1, \dots, A_{k+1} \rangle$ (so assume only one axiom A_{k+1} is missing) and let Q be a consequent of J . Now treat A_{k+1} as a missing axiom and let I be the ideal generated by the known axioms and the consequence $\langle A_1, \dots, A_k, Q \rangle$. If I and J have the same height, then we know that they have a common associated prime which must contain the missing axiom A_{k+1} . Under the usual situation where Q and A_{k+1} are not contained in any of the associated primes for $\langle A_1, \dots, A_k \rangle$, then we have that $\text{ht}\langle A_1, \dots, A_k, Q \rangle = \text{ht}\langle A_1, \dots, A_{k+1} \rangle$, the theorem applies, and we can examine the associated primes for I to recover the missing axiom A_{k+1} .

Geometrically, each axiom defines a hypersurface that reduces the dimension of the solution space by at most one. Removing an axiom removes a cut, increasing the variety’s dimension. Adding a hypothesis Q introduces a new cut. The theorem states that if the dimension lost by removing

axioms is exactly recovered by adding hypotheses, then the decomposition of I will reveal a component containing the missing axiom. In other words, the hypotheses carry enough information to reconstruct the missing constraints.

Typically, if multiple axioms are missing, adding back a single phenomenon Q will not be sufficient, so in general we will need to add back several consequent phenomena $\{Q_1, \dots, Q_h\}$ in order to generate an ideal which is guaranteed to have a component that contains all the missing axioms.

Recovering Axioms With Noise in the Consequences

When the consequences are noisy, then it is no longer true that removing an axiom and adding a noisy consequent produces an ideal that is exactly contained in the original axiom ideal. It is in some sense close to an ideal which is so contained. To handle this, we replace exact ideal decomposition with numerical ideal decomposition [3]. This produces sample points on each component of the perturbed ideal. Assuming the perturbation is small and generic, we expect sample points for each numerical component to be approximate zeros of one of the components of the true ideal [8]. When this is the case we can use symbolic regression on the sample points generated by numerical decomposition to generate approximations to the missing axiom.

Comparison with State-of-the-Art

We evaluate our abductive reasoning system against four baselines: 1) Traditional Symbolic Regression (e.g., AI-Feynman [48], Eureka [15], and PySR [13]), which fits a model directly to observational data; 2) Integrated Symbolic Regression (e.g., AI-Hilbert [11] and AI-Descartes [9]), which fits data while enforcing consistency with background theory; 3) LLM-based abduction using GPT-5 Pro constrained to output polynomial hypotheses; and 4) solver-based abduction using cvc5 with SyGuS grammars. All methods are tested on a common benchmark of algebraic physics systems, where one or more axioms are removed, and the goal is to infer a missing axiom that will provide derivability of a target conjecture. Each proposed axiom set is checked by the same symbolic verification pipeline in Macaulay2 (algebraic verifier). Our experiments are both on clean and noisy data for each problem in our benchmark set, and use multiple levels of axiom removal.

Overall, our system demonstrates substantially stronger performance than all four baselines (see Table 4). Both symbolic-regression baselines require observational data for the relationship being fit, whereas we assume that no data are available for the missing axioms; Traditional Symbolic Regression and AI-Descartes therefore cannot be applied in our abductive setting. AI-Hilbert can instead be run without data in a purely deductive mode, attempting to derive the missing axiom from the known axioms and consequence, but this fails in every case, since by construction, and in our benchmarks in reality, the missing axioms are not deductively derivable from the known axioms and consequences. Consequently, only the LLM and cvc5 baselines operate in our setting, and these are evaluated on the same benchmark. AI-Noether succeeds in 97% of single-axiom cases where the two data-free baselines achieved 0% (LLM) or timed out (cvc5), including cases where the missing axiom involves numerical constants or requires longer multi-step symbolic reasoning. cvc5 generally fails to generate a polynomial within the allowed grammar, as it exceeds the runtime we set. GPT-5 Pro can generate syntactically reasonable expressions but doesn’t yield axioms that satisfy the strict algebraic membership or literal-appearance criteria. Instead, our approach, in most cases, recovers correct and theory-compatible axioms and remains robust to noise.

More details about our experiments comparing AI-Noether with the state of the art can be found in Supplementary Note 6.

Data Availability

The benchmark axiom systems (nine of which were originally reported in AI-Hilbert [11]) and the synthetic datasets generated in this study are available at the AI-Noether GitHub repository [47] (<https://github.com/IBM/AI-Noether>) without restriction.

Code Availability

The code used for this work can be found, freely available, at the AI-Noether GitHub repository [47]: <https://github.com/IBM/AI-Noether>. The code is released under the MIT License. Algorithm implementation details are provided in Supplementary Note 5. The repository includes documentation for reproducing all experiments presented in this paper.

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Author Contributions Statement

K.S. conceptualized and developed the overall abductive inference framework, wrote the codebase and prepared it for release, designed and ran experiments, helped set up AI Hilbert and the polynomial optimization framework for numerical experiments, analyzed the algebraic and numerical results to validate the framework, generated figures, and co-authored the manuscript. S.D. contributed to the conceptualization of the overall abductive inference framework, designed some of the experiments, analyzed results to validate the framework, helped set up AI Hilbert and the polynomial optimization framework for numerical experiments, and co-authored the manuscript. R.C.W. contributed to the conceptualization of the abductive inference framework, helped set up AI Hilbert and the polynomial optimization framework for numerical experiments, analyzed the validity of the numerical results, designed some of the experiments, and co-authored the manuscript. B.T. contributed to the conceptualization of the abductive inference framework, derived proofs of correctness and validation for the numerical framework, designed experiments, and helped analyze the results to validate the framework, and co-authored the manuscript. C.C. contributed to the conceptualization of the abductive inference framework, designed the formal logic verification framework, designed the code for formal logic verification, analyzed results to validate the framework, generated figures, and co-authored the manuscript. L.H. contributed to the conceptualization of the abductive inference framework and the development of the numerical nonlinear framework, scoped and led the broader Science 2.0 program seeking to transform the scientific method with AI and mathematics of which this work was a part, designed some of the experiments, analyzed the results to validate the framework, and co-authored the manuscript.

Competing Interests Statement

The authors declare no competing interests.

Tables

Problem	Axioms	Success Rate	Variables	Max Degree
Hagen-Poiseuille Equation	4	100%	9	4
Kepler	5	100%	8	4
Time Dilation	5	100%	8	4
Escape Velocity	5	100%	9	4
Light Damping	5	100%	10	8
Neutrino Decay	5	100%	8	2
Simple Harmonic Oscillator	6	100%	11	5
Carrier-Resolved Photo-Hall	7	100%	15	6
Relativistic Laws	7	71%*	12	5
Hall Effect	7	100%	16	4
Inelastic Relativistic Collision	9	100%	11	6
Compton Scattering	10	100%	15	6

Table 1: Summary of Axiom Recovery Performance by Problem. In each case, we iterate through every axiom, remove it from the theory, and attempt to recover the axiom with the remaining information. In almost every case, we are able to recover every axiom. *Indicates partial recovery due to recovering alternate axioms. We discuss this particular case further in the Multiple Consequences subsection of the Methods.

Pairs of missing axioms		Triples of missing axioms	
Missing Axioms	Recovered	Missing Axioms	Recovered
{G, cf}	X	{G, cf, C}	X
{G, C}	✓	{G, cf, wp}	X
{G, wp}	X	{G, cf, cm}	X
{G, cm}	✓	{G, C, wp}	X
{cf, C}	✓	{G, C, cm}	✓
{cf, wp}	✓	{G, wp, cm}	X
{cf, cm}	X	{cf, C, wp}	✓
{C, wp}	✓	{cf, C, cm}	X
{C, cm}	✓	{cf, wp, cm}	X
{wp, cm}	✓	{C, wp, cm}	X

Table 2: Recovery results on the Kepler problem for various subsets of missing axioms. Shorthand: G = gravitational force $(d_1 + d_2)^2 F_g = m_1 m_2$, cf = centrifugal force $F_c = m_2 d_2 \omega^2$, C = force balance $F_g = F_c$, wp = frequency-period relation $\omega p = 1$, cm = center of mass $m_1 d_1 = m_2 d_2$. ✓ indicates that the missing axioms were recovered; X indicates that they were not.

Problem	Vars	Axioms	Pairs	Triples	2-Ax Success	3-Ax Success
Hagen-Poiseuille	9	4	6	4	3/6	1/4
Kepler	8	5	10	10	7/10	2/10
Time Dilation	8	5	10	10	1/10	1/10
Escape Velocity	9	5	10	10	8/10	4/10
Light Damping	10	5	10	10	1/10	0/10
Neutrino Decay	8	5	10	10	3/10	1/10
Simple Harmonic Osc.	11	6	15	20	10/15	4/20
Carrier-Resolved PH	15	7	21	35	13/21	15/35
Relativistic Laws	12	7	21	35	17/21	24/35
Hall Effect	16	7	21	35	13/21	17/35
Inelastic Relativistic Collision	11	9	36	84	12/36	15/84
Compton Scattering	15	10	45	120	17/45	25/120

Table 3: Summary of 2- and 3-Axiom Tuple Recovery Performance by Problem. Pairs = $\binom{n}{2}$, Triples = $\binom{n}{3}$ where n is the number of axioms. Vars = number of variables; 2-Ax Success and 3-Ax Success = number of recovered axiom pairs and triples, respectively; Carrier-Resolved PH = Carrier-Resolved Photo-Hall; Simple Harmonic Osc. = Simple Harmonic Oscillator.

Method	Single Axiom	Multi Axiom	Data-Free	Noise Robust	Var. Agnostic	Explainable
Traditional SR	✗	✗	✗	✗	✗	✓
Integrated SR	✗	✗	✗	✓	✓	✓
LLM (GPT-5 Pro)	0%	0%	✓	✗	✓	✓
cvc5	0%*	0%*	✓	✗	✗	✓
AI-Noether	97%	49%	✓	✓	✓	✓

Table 4: **Comparison of abductive and symbolic regression methods.** Ticks (✓) indicate capability support; crosses (✗) indicate absence. Data-Free means the method operates without numeric data or prior parameter information, and Var. Agnostic indicates invariance to variable naming. SR = symbolic regression. Percentages in the first two columns indicate the percentage of cases that successfully recovered across all 12 problems reported on in the Single Missing Axiom through Noisy Consequences subsections of the Methods.

* = due to timeout, meaning that the method ran out of time (set to 12 hours on the hardware detailed in Supplementary Note 5) without producing a solution.

Figure Legends/Captions

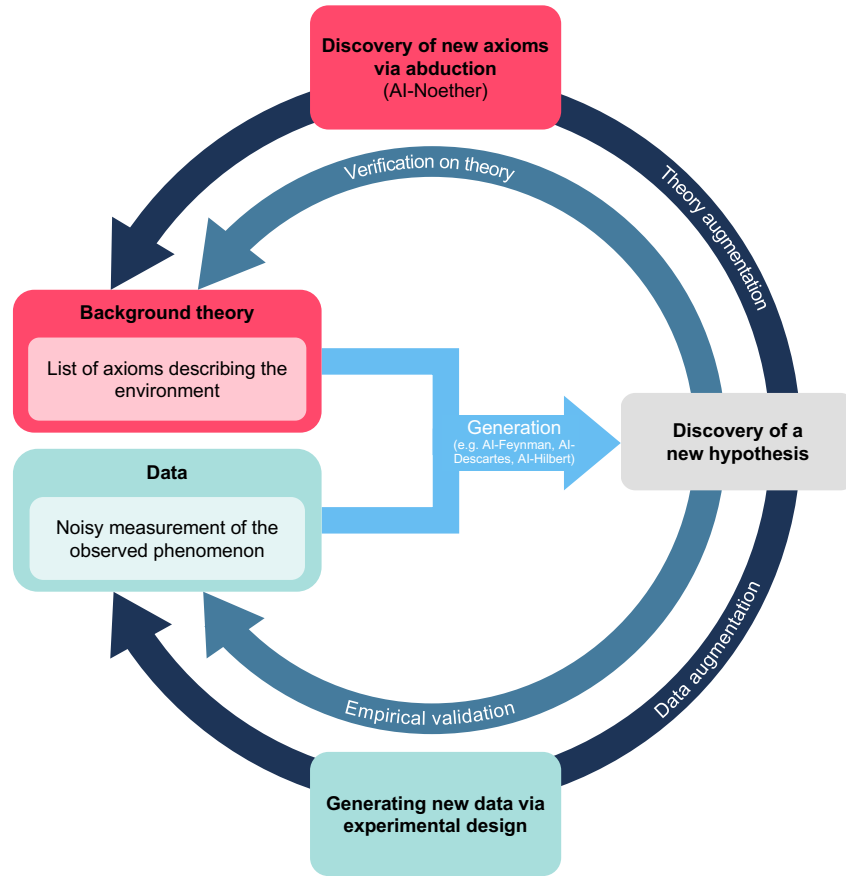


Figure 1: AI-Noether-augmented scientific discovery loop. Background theory axioms and observational data may give rise to new candidate hypotheses that may fit the data but fail to be derivable from existing theory. AI-Noether acts on this theory-hypothesis pair: it performs abductive inference to identify missing axioms that reconcile these hypotheses with canonical knowledge, while symmetrically, on the data-driven end, experimental design prescribes the generation of new data. Both procedures are instrumental to further refine both hypotheses and axioms, closing the discovery-verification loop. Red boxes denote theory-side components, teal boxes denote data-side components, and the grey box denotes the new hypothesis; dark blue arrows denote augmentation, medium blue arrows denote verification and validation, and the light blue arrow denotes hypothesis generation.

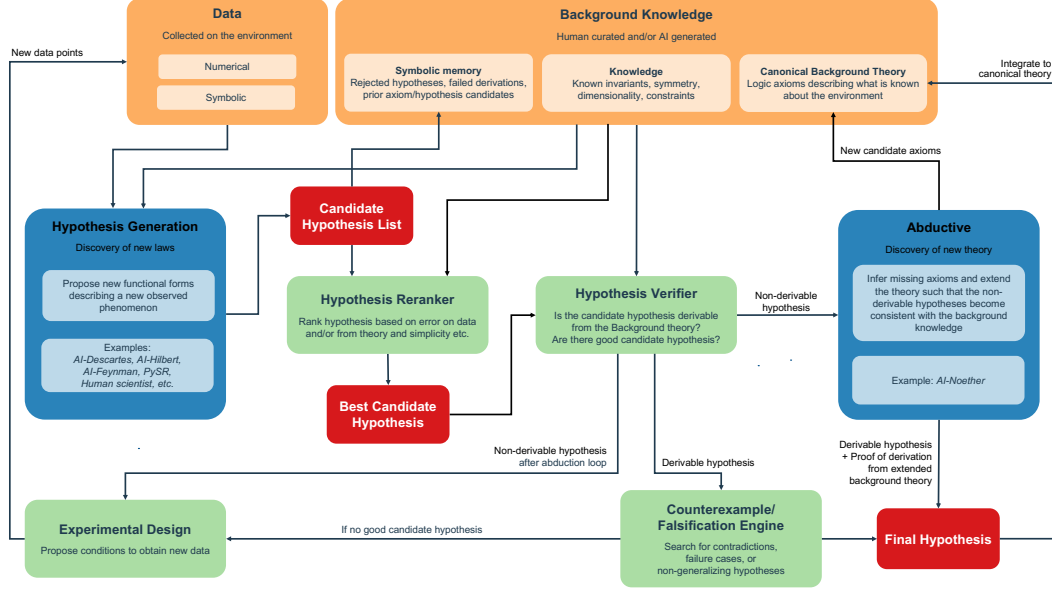


Figure 2: **AI-Noether inside a broader scientific discovery loop.** Data and background knowledge support the generation, reranking, verification, and falsification of candidate hypotheses. When a hypothesis is not derivable from the current background theory, AI-Noether performs abductive discovery by inferring missing axioms that make the hypothesis derivable. The resulting axioms can be integrated into the background theory, while experimental design and symbolic memory support further iterations of the loop. Orange boxes denote data and background knowledge, blue boxes denote hypothesis-generation and abductive modules, green boxes denote reranking, verification, falsification and experimental-design modules, and red boxes denote candidate and final hypotheses.

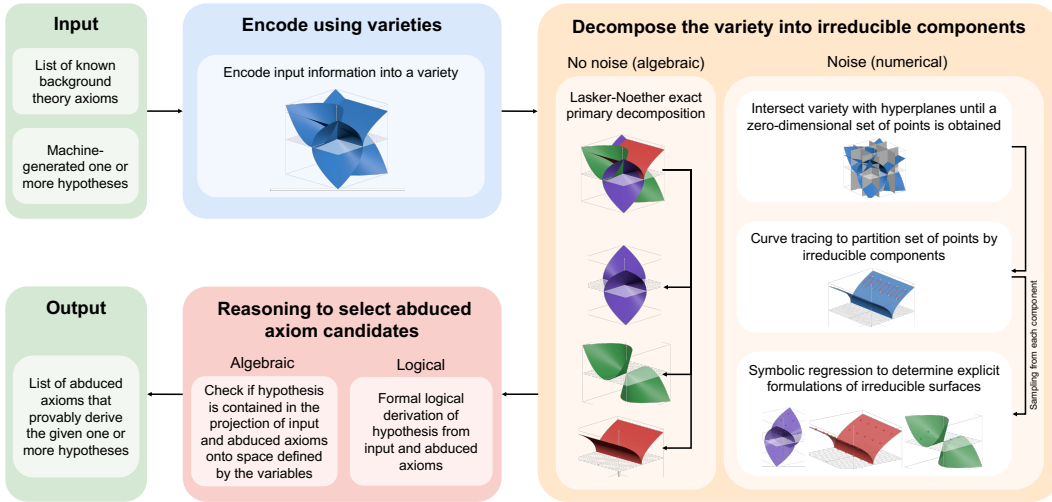


Figure 3: **AI-Noether system overview.** Given background theory axioms and one or more candidate hypotheses that are not supported by the existing theory, AI-Noether identifies additional axioms that reconcile the hypotheses with the background knowledge. The system first encodes axioms and hypotheses into a geometric representation, decomposes this representation into fundamental components, and then reasons over these components to determine which candidate axioms, when added to the theory, are sufficient to explain the hypotheses. These axioms are returned as output after verification, which proves the hypotheses, using both algebraic and logic-based methods. Green panels denote input and output, and the blue, orange and red panels denote the encode, decompose and reason steps, respectively; within the panels, coloured surfaces depict varieties and their irreducible components.

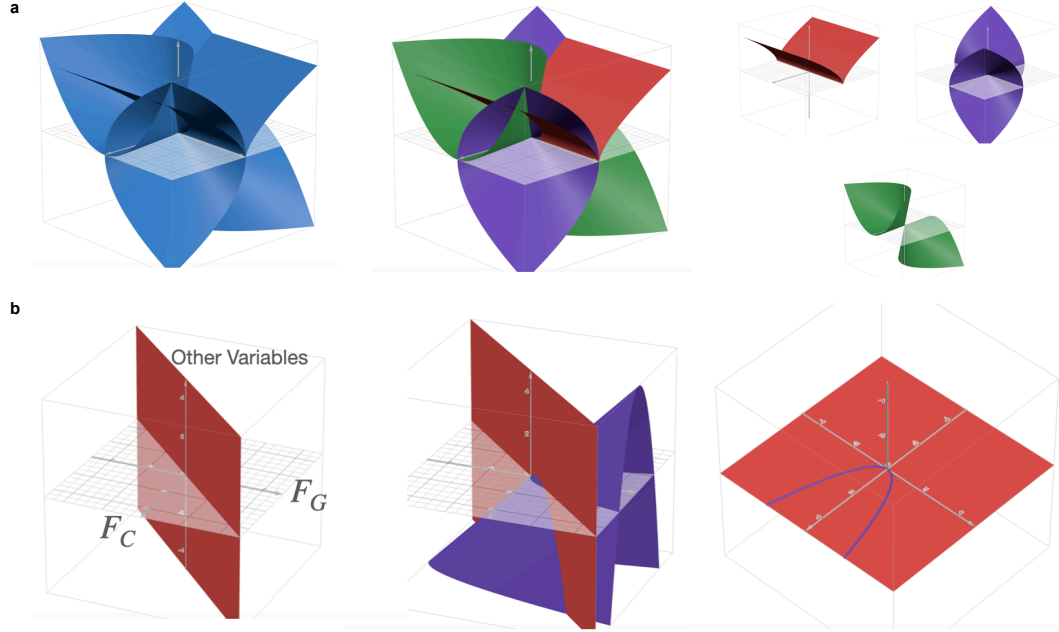


Figure 4: **Geometric visualization of decomposition and algebraic reasoning.** **a**, On the left, we have an example of a variety which is the solution to a system of polynomial equations. The middle and right panels show that this variety can be decomposed into irreducible components, which, geometrically, is the process of finding smaller varieties whose union is the original variety. The full variety is shown in blue; each irreducible component is shown in a distinct colour (red, green, purple). **b**, On the left, we have some surface defined by an axiom (here, $F_C = F_G$ as an example, where F_C and F_G are the centrifugal and gravitational forces, written F_c and F_g in the text), shown in red. In the middle, we have an additional equation that defines a new surface, shown in purple. On the right, in order to apply the axiom to the new surface, we project the surface onto the space defined by the known axiom(s).

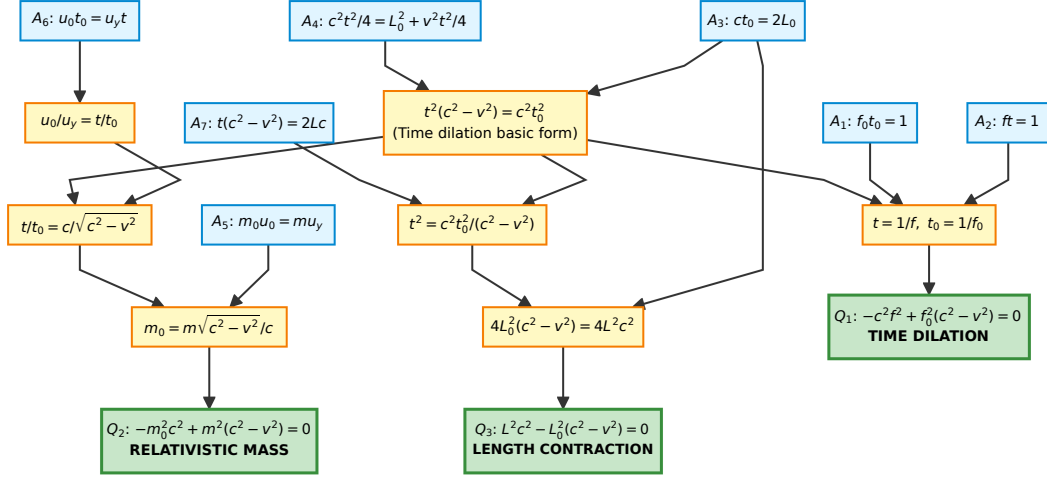


Figure 5: **Derivation structure for relativistic consequences.** The graph shows how axioms A_1 - A_7 combine through intermediate results to derive the three consequences. Solid blue boxes represent axioms, yellow boxes show intermediate algebraic steps, and green boxes are the final consequences. The derivation of Q_1 (time dilation) uses axioms A_1 - A_4 ; Q_2 (relativistic mass) also requires A_5 - A_6 ; and Q_3 (length contraction) further needs A_7 . The intermediate steps, indicated in yellow, are a manual illustration of one derivation of Q_1 , Q_2 , and Q_3 . f_0 and f , frequencies in the rest and moving frames; t_0 and t , the corresponding periods; c , speed of light; v , velocity of the moving frame; L_0 and L , lengths in the rest and moving frames; m_0 and m , rest and relativistic mass; u_0 and u_y , transverse velocities in the rest and moving frames.

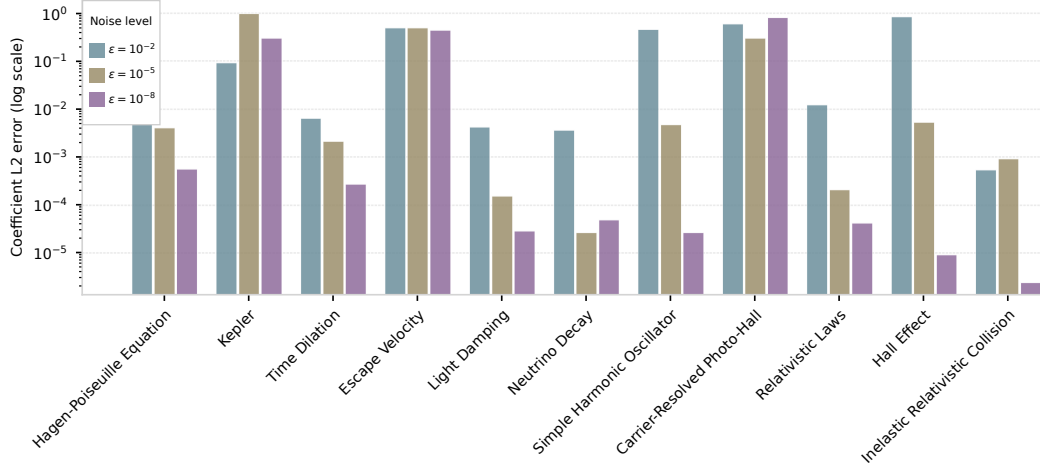


Figure 6: **Results on running AI-Noether on various problems** with 1, 2, and 3 axioms missing with noisy consequences obtained from running AI-Hilbert [11] with varying levels of noise in the data. For each problem, we plot the normalized coefficient distance [11, 12] and the noise level for successful runs. For some problems, the error is non-monotone in ϵ ; we diagnose this behavior in Supplementary Note 5, where we isolate AI-Noether's noise sensitivity from the contribution of the upstream AI-Hilbert step. The normalized coefficient distance is labelled Coefficient L2 error (log scale) on the vertical axis.